```
# ============================================================
#  ZeroG Mercury One.1 (Ender 5 CoreXY conversion)
#  BTT Octopus Pro v1.0 · Triple independent Z
#  Sherpa Mini · Cartographer V4 · Sensorless XY homing
# ============================================================
#
#  PIN SOURCE: Klipper's generic-bigtreetech-octopus-pro-v1.0.cfg
#  (repo @ 2026-08-25). Every board pin below is copied from it.
#
#  >>> BOARD REVISION: Octopus Pro v1.0 <<<
#  This file is now set for a Pro **v1.0**. Two pins differ from the
#  v1.1 wiring and are already changed below:
#     HE0 heater      PA2  (v1.1 uses PA0)
#     Driver3 enable  !PA0 (v1.1 uses !PA2)
#  Do NOT use this file on a v1.1 board - HE0 and the Driver3 enable
#  line swap, which can enable a heater unintentionally.
#
#  A Pro v1.0 may carry an STM32F446, STM32F429 *or* STM32H723.
#  Read the chip and match "make menuconfig" to it - see the guide.
#
#  >>> SENSORLESS HOMING — WIRING PREREQUISITES <<<
#  1. Fit the DIAG jumper on the Driver 0 and Driver 1 sockets.
#  2. UNPLUG any physical X/Y endstop switches. They share PG6/PG9
#     with the DIAG lines and will hold the pin permanently triggered.
#  3. Never set hold_current on stepper_x or stepper_y — it breaks
#     stall detection after contact is made.
#
#  Lines marked  ### VERIFY  are starting points, not measurements.
# ============================================================


# ------------------------------------------------------------
# MCU
# ------------------------------------------------------------
[mcu]
serial: /dev/serial/by-id/usb-Klipper_stm32f446xx_XXXXXXXXXXXX-if00   ### VERIFY
# Get the real path with:  ls /dev/serial/by-id/
restart_method: command


# ------------------------------------------------------------
# Cartographer V4 probe (its own MCU)
# ------------------------------------------------------------
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[mcu cartographer]
serial: /dev/serial/by-id/usb-Cartographer_XXXXXXXX-if00          ### VERIFY
# CAN instead? Delete serial: and use  canbus_uuid: <uuid>
restart_method: command

[cartographer]
mcu: cartographer
# MEASURE THESE OFF YOUR TOOLHEAD CAD. Both zeros is almost certainly
# wrong and will shift your whole bed mesh. Coil centre minus nozzle.
x_offset: 0               ### VERIFY
y_offset: 0               ### VERIFY
verbose: no

[temperature_sensor cartographer_mcu]
sensor_type: temperature_mcu
sensor_mcu: cartographer

[adxl345]
cs_pin: cartographer:PA3      ### VERIFY against your Cartographer revision
spi_bus: spi1

[resonance_tester]
accel_chip: adxl345
probe_points:
   117.5, 117.5, 20


# ------------------------------------------------------------
# Kinematics
# ------------------------------------------------------------
[printer]
kinematics: corexy
max_velocity: 300          # conservative for commissioning
max_accel: 3000            # raise after input shaping
max_z_velocity: 15         # TR8x8 confirmed. 15mm/s = 112 rpm, safe;
                           # can go ~25 once the Z chain is proven.
max_z_accel: 100
square_corner_velocity: 5.0


# ------------------------------------------------------------
# CoreXY motors — Driver 0 (A) and Driver 1 (B)
# ------------------------------------------------------------
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```
# Mercury One.1 homes X and Y to MAXIMUM (rear-right corner).
# Neither stepper drives a single axis on its own.
#
# Sensorless: the endstop is the driver's own stall detection, so
# endstop_pin points at a virtual endstop and homing_retract_dist
# MUST be 0 (the second homing touch is unreliable when sensorless).
#
# homing_speed 20 = rotation_distance / 2, which is Klipper's
# recommended starting point: roughly one motor rev every two
# seconds. Too slow and there is no back-EMF to detect a stall.

[stepper_x]
step_pin: PF13
dir_pin: PF12                # add/remove leading ! to reverse
enable_pin: !PF14
microsteps: 32
rotation_distance: 40        # 2GT belt, 20-tooth pulley
full_steps_per_rotation: 200  # 1.8° steppers
endstop_pin: tmc2209_stepper_x:virtual_endstop
position_endstop: 235        ### VERIFY — measure your actual X travel
position_max: 235            ### VERIFY
position_min: 0
homing_positive_dir: true
homing_speed: 20
homing_retract_dist: 0       # required for sensorless

[stepper_y]
step_pin: PG0
dir_pin: PG1
enable_pin: !PF15
microsteps: 32
rotation_distance: 40
full_steps_per_rotation: 200
endstop_pin: tmc2209_stepper_y:virtual_endstop
position_endstop: 235        ### VERIFY
position_max: 235            ### VERIFY
position_min: 0
homing_positive_dir: true
homing_speed: 20
homing_retract_dist: 0

# driver_SGTHRS: 255 = most sensitive, 0 = least. 100 is a placeholder
# — you WILL tune this in Stage 5. No hold_current, deliberately.
```

```
[tmc2209 stepper_x]
uart_pin: PC4
diag_pin: ^PG6              # DIAG0 — needs the jumper fitted
driver_SGTHRS: 100          ### VERIFY — tuned, not guessed
run_current: 0.800          ### VERIFY — ~0.7 × motor rated RMS
stealthchop_threshold: 0    # spreadCycle; does not affect sensorless
interpolate: False

[tmc2209 stepper_y]
uart_pin: PD11
diag_pin: ^PG9              # DIAG1 — needs the jumper fitted
driver_SGTHRS: 100          ### VERIFY
run_current: 0.800          ### VERIFY
stealthchop_threshold: 0
interpolate: False


# ------------------------------------------------------------
# Triple independent Z — Drivers 2, 3, 5
# ------------------------------------------------------------
# Driver 4 is the extruder, so the third Z lives on Driver 5.
# Z homing is done by the Cartographer — no physical Z endstop.

# Driver 2 — front left leadscrew        ### VERIFY which screw is which
[stepper_z]
step_pin: PF11
dir_pin: PG3
enable_pin: !PG5
microsteps: 32
rotation_distance: 8        # TR8x8 confirmed (2mm pitch x 4 starts)
full_steps_per_rotation: 200
endstop_pin: probe:z_virtual_endstop
homing_retract_dist: 0      # required by Cartographer
position_max: 300           ### VERIFY
position_min: -5

# Driver 3 — front right leadscrew
[stepper_z1]
step_pin: PG4
dir_pin: PC1
enable_pin: !PA0            # Driver3 on Pro v1.0 (!PA2 on v1.1)
microsteps: 32
rotation_distance: 8        # TR8x8 confirmed
```

```
full_steps_per_rotation: 200

# Driver 5 — rear centre leadscrew
[stepper_z2]
step_pin: PC13
dir_pin: PF0
enable_pin: !PF1
microsteps: 32
rotation_distance: 8          # TR8x8 confirmed
full_steps_per_rotation: 200

[tmc2208 stepper_z]
uart_pin: PC6
run_current: 0.600            ### VERIFY
stealthchop_threshold: 999999

[tmc2208 stepper_z1]
uart_pin: PC7
run_current: 0.600
stealthchop_threshold: 999999

[tmc2208 stepper_z2]
uart_pin: PE4
run_current: 0.600
stealthchop_threshold: 999999

# Order below MUST match stepper_z, stepper_z1, stepper_z2.
# z_positions are where each leadscrew meets the bed, in nozzle
# coordinates; they may fall outside the bed. This front-left /
# front-right / rear-centre layout follows the published Mercury
# One.1 triple-Z reference builds. Measure yours.
[z_tilt]
z_positions:
    10, 0                   ### VERIFY — front left screw
    225, 0                  ### VERIFY — front right screw
    117.5, 210              ### VERIFY — rear centre screw
points:
    30, 30                  ### VERIFY — reachable AND on the bed
    205, 30
    117.5, 195
speed: 150
horizontal_move_z: 5
retries: 8
```

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retry_tolerance: 0.0075


# ------------------------------------------------------------
# Extruder — Driver 4, Sherpa Mini
# ------------------------------------------------------------
[extruder]
step_pin: PF9
dir_pin: PF10
enable_pin: !PG2
microsteps: 32
rotation_distance: 22.6789511 ### VERIFY — starting value, MUST be calibrated
gear_ratio: 50:10            # Sherpa Mini 5:1
full_steps_per_rotation: 200  ### VERIFY — 400 if you fitted a 0.9° pancake
nozzle_diameter: 0.400        ### VERIFY
filament_diameter: 1.750
heater_pin: PA2              # HE0 on Pro v1.0 (PA0 on v1.1)
sensor_pin: PF4               # T0
sensor_type: Generic 3950     ### VERIFY — match your actual thermistor
min_temp: 0
max_temp: 300                 ### VERIFY — 250 for a PTFE-lined hotend
max_extrude_only_distance: 150
pressure_advance: 0.0         # calibrated later
pressure_advance_smooth_time: 0.040
control: pid
pid_Kp: 22.2                 # placeholders — run PID_CALIBRATE
pid_Ki: 1.08
pid_Kd: 114

[tmc2209 extruder]
uart_pin: PF2
run_current: 0.550           ### VERIFY — pancake steppers are low-current
stealthchop_threshold: 0


# ------------------------------------------------------------
# Bed
# ------------------------------------------------------------
[heater_bed]
heater_pin: PA1
sensor_pin: PF3              # TB
sensor_type: Generic 3950     ### VERIFY
min_temp: 0
```

```
max_temp: 120
control: pid
pid_Kp: 54.0
pid_Ki: 0.77
pid_Kd: 950


# ------------------------------------------------------------
# Fans
# ------------------------------------------------------------
[fan]                       # part cooling — FAN0
pin: PA8

[heater_fan hotend_fan]       # FAN1
pin: PE5
heater: extruder
heater_temp: 50.0

[controller_fan board_fan]    # FAN2 — cools the stepper drivers
pin: PD12
stepper: stepper_x, stepper_y, stepper_z


# ------------------------------------------------------------
# Bed mesh
# ------------------------------------------------------------
[bed_mesh]
speed: 200
horizontal_move_z: 5
mesh_min: 25, 25              ### VERIFY — must clear the probe offsets
mesh_max: 210, 190            ### VERIFY — Mercury Ender 5 reference value
probe_count: 15, 15
algorithm: bicubic
zero_reference_position: 117.5, 117.5   # bed centre — required for touch


# ------------------------------------------------------------
# Homing
# ------------------------------------------------------------
# No [safe_z_home] — homing_override replaces it entirely. The two
# cannot coexist. This override still satisfies Cartographer's
# requirement that G28 finish homing Z at the centre of the bed.
#
```

```
# Sequence matters on CoreXY sensorless: home one axis, move it off
# the rail limit, wait for the driver's stall flag to clear, then
# home the other. Homing X while Y is jammed against its limit skews
# the stall reading.

[homing_override]
axes: xyz
gcode:
   {% set home_all = 'X' not in params and 'Y' not in params and 'Z' not in params %}

   # Get the nozzle clear of the bed before any XY motion.
   # SET_HOMED=Z is required: a bare SET_KINEMATIC_POSITION marks
   # ALL axes homed, which would silently skip the real X/Y homing.
   {% if 'z' not in printer.toolhead.homed_axes %}
      SET_KINEMATIC_POSITION Z=0 SET_HOMED=Z
   {% endif %}
   G90
   G0 Z10 F900

   {% if home_all or 'Y' in params %}
      _SENSORLESS_HOME_Y
   {% endif %}

   {% if home_all or 'X' in params %}
      _SENSORLESS_HOME_X
   {% endif %}

   {% if home_all or 'Z' in params %}
      # A bare "G28 Z" needs X and Y known first — the Cartographer
      # has to probe at bed centre, and that move needs valid XY.
      {% if 'y' not in printer.toolhead.homed_axes %}
         _SENSORLESS_HOME_Y
      {% endif %}
      {% if 'x' not in printer.toolhead.homed_axes %}
         _SENSORLESS_HOME_X
      {% endif %}
      G90
      G0 X117.5 Y117.5 F9000
      G28 Z
      G0 Z10 F900
   {% endif %}

[gcode_macro _SENSORLESS_HOME_X]
```

```
gcode:
   {% set HOME_CUR = 0.500 %}
   {% set RUN_CUR = printer.configfile.settings['tmc2209 stepper_x'].run_current %}
   # Both CoreXY motors move during an X home, so drop both currents.
   SET_TMC_CURRENT STEPPER=stepper_x CURRENT={HOME_CUR}
   SET_TMC_CURRENT STEPPER=stepper_y CURRENT={HOME_CUR}
   G4 P2000                         # let the stall flag clear
   G28 X0
   G90
   G0 X{printer.toolhead.axis_maximum.x - 10} F1800
   G4 P500
   SET_TMC_CURRENT STEPPER=stepper_x CURRENT={RUN_CUR}
   SET_TMC_CURRENT STEPPER=stepper_y CURRENT={RUN_CUR}

[gcode_macro _SENSORLESS_HOME_Y]
gcode:
   {% set HOME_CUR = 0.500 %}
   {% set RUN_CUR = printer.configfile.settings['tmc2209 stepper_y'].run_current %}
   SET_TMC_CURRENT STEPPER=stepper_x CURRENT={HOME_CUR}
   SET_TMC_CURRENT STEPPER=stepper_y CURRENT={HOME_CUR}
   G4 P2000
   G28 Y0
   G90
   G0 Y{printer.toolhead.axis_maximum.y - 10} F1800
   G4 P500
   SET_TMC_CURRENT STEPPER=stepper_x CURRENT={RUN_CUR}
   SET_TMC_CURRENT STEPPER=stepper_y CURRENT={RUN_CUR}


# ------------------------------------------------------------
# Misc
# ------------------------------------------------------------
[idle_timeout]
timeout: 1800

[exclude_object]

[force_move]
enable_force_move: True       # lets you nudge one Z motor during setup


# ------------------------------------------------------------
# Macros
```

```
# ------------------------------------------------------------
[gcode_macro G32]
gcode:
    G28
    Z_TILT_ADJUST
    G28 Z
    G0 X117.5 Y117.5 Z20 F6000

[gcode_macro PRINT_START]
gcode:
    {% set BED = params.BED|default(60)|float %}
    {% set EXTRUDER = params.EXTRUDER|default(200)|float %}
    M140 S{BED}
    M104 S150                   ; safe probing temp, no ooze
    M190 S{BED}
    G28
    Z_TILT_ADJUST
    G28 Z
    BED_MESH_CALIBRATE ADAPTIVE=1
    G0 X117.5 Y117.5 Z20 F6000
    M109 S{EXTRUDER}
    G90
    G92 E0

[gcode_macro PRINT_END]
gcode:
    M400
    G92 E0
    G1 E-5.0 F1800
    TURN_OFF_HEATERS
    M107
    G91
    G1 Z10 F3000
    G90
    G0 X117.5 Y225 F6000
    M84
    BED_MESH_CLEAR


# ------------------------------------------------------------
# Added later by calibration — do not hand-edit below this line
# ------------------------------------------------------------
# [input_shaper]  ← added after SHAPER_CALIBRATE
```

#*# <---------------------- SAVE_CONFIG ---------------------->